Summary of all protocols used in wet lab

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Plasmid extraction

E. coli DH10-beta cells carrying pET28, pET29 and pBEVY plasmids were grown overnight in 3 mL Luria-Bertani (LB) broth medium with 100 µg/mL ampicillin or 50 µg/mL kanamycin. Less than 18 hours after the inoculation of the overnight liquid cultures, the cells were pelleted by centrifugation at 4000 g for 10 minutes. Upon isolation of the cell pellet, plasmids were isolated according to the protocol from the GeneJET Plasmid Miniprep Kit (ThermoFischer Scientific), according to the manufacturer's protocol as well.

Primer design and polymerase chain reaction (PCR)

Primers to amplify gene block (gBlocks, IDT) coding for proteins were designed using the Benchling Assembly Wizard and SnapGene. Two types of constructs were designed: Gibson Assembly and Golden Gate-based assembled vectors. For the Golden Gate-based assemblies, BsaI sites were included by PCR using the primer sequences shown below. For the Gibson Assembly-derived constructs, a DNA sequence coding for a flexible polypeptide linker (SGGSGGS) between the recombinant protein and sfGFP was introduced using PCR.

gBlocks with DNA sequences coding for biosurfactant proteins were amplified using a DNA Taq polymerase (NEB). Plasmid vector backbones were amplified using the 2X Q5 Master Mix (NEB). The Amplification conditions were the following. PCR products amplified from plasmid templates were digested using Dpn1 for 1 hour at 37°C. All PCR products were column-purified using the GeneJET PCR Purification Kit (Thermo Fischer Scientific) and were run on agarose gels to verify the formation of a PCR product of the appropriate length. Table 1 shows the primers used for the successfully cloned E. coli constructs.

Table 1: primers used for the successfully cloned E. coli constructs.

Primer name	**Primer sequence (5'-3')**
pET29_Gibs_fwd	AGTGGAGGTTCTGGAGGTAGCATGGTTAGCAAAGGTGAAG
pET29_Gibs_rev	TGAAGTGTATATCTCCTTCttaaag
MBSP1_Gibs_fwd	AAGAAGGAGATATACACTTCAATGTCGGACCAATACCTTGACTTtgg
MBSP1_Gibs_rev	GCTACCTCCAGAACCTCCACTCGTGGAATCCTGGCCCG
HFBI_Gibs_fwd	aaGAAGGAGATATACACTTCAATGTCTAACGGCAATGGTAACG
HFBI_Gibs_rev	GCTACCTCCAGAACCTCCACTCGCCCCAACCGCTG
HFBI_mut_Gibs_fwd	aaGAAGGAGATATACACTTCAATGTCCAACGGTAATGGCAACG

HFBI_mut_Gibs_rev	GCTACCTCCAGAACCTCCACTTGCCC CCACCGCTG

A complete spreadsheet of all primers used throughout the project can be found in annex one attached to this document.

Q5 polymerase

Vector backbones were amplified using the Q5® High-Fidelity 2X Master Mix (NEB) according to the manufacturer's specifications, in 50 µL reactions and with 10 ng of plasmid template per reaction. The annealing temperatures were calculated using the NEB Tm Calculator tool. The cycling parameters are shown in table 2.

Table 2: PCR cycling parameters used to amplify vector backbones.

STEP	TEMP	TIME
Initial Denaturation	98°C	30 seconds
30 Cycles	98°C	10 seconds
	Annealing Temperature	30 seconds
	72°C	3 minutes 30 seconds
Final Extension	72°C	2 minutes
Hold	16°C	

Taq polymerase

5 ng of gBlock DNA coding for the desired inserts was amplified using Taq DNA polymerase to make it compatible with Type IIs/Gibson Assembly cloning strategies. Reactions were set-up using NEB's Taq DNA polymerase. The cycling parameters that were used are shown in table 3.

Table 3: PCR cycling parameters used for the desired inserts.

STEP	TEMP	TIME
Initial Denaturation	95°C	30 seconds
30 Cycles	95°C	30 seconds
	Annealing Temperature from NEB Tm Calculator	60 seconds
	68°C	20 seconds
Final Extension	68°C	5 minutes
Hold	16°C	

Annealing and phosphorylation of oligonucleotides

Oligonucleotides (IDT) were also used in cloning to introduce a short polypeptide linker upstream of the sfGFP sequence on the plasmid. First, the oligonucleotides were treated separately with T4 Polynucleotide Kinase (NEB) in the presence of 10 XT4 DNA ligase buffer (reference). The mixture was incubated at 37 °C for 1 hour then heat inactivated for 20 minutes at 65 °C. Following the phosphorylation reaction, the oligonucleotides were annealed into a duplex with overhangs compatible with Type IIS cloning. Annealing was done by mixing the oligonucleotides in equal molar ratios and incubating at 95 °C for 3 minutes, followed by slow cooling to room temperature. The concentration of the annealed oligonucleotide duplex was assessed using a spectrophotometer.

Type IIS cloning

Constructs were designed using the Assembly Wizard in Benchling. A designed modular construct was used, such that the same vector backbone can be used to clone multiple fragments, which would all be compatible with the same expression platform. An intermediate cloning vector was also derived from the original pET29-sfGFP plasmid, which includes a 21 bp linker upstream of the sfGFP sequence. This pET29-linker-sfGFP plasmid was constructed using a PCR-amplified vector backbone and an annealed and phosphorylated oligonucleotide duplex. PCR products containing BsaI sites were digested using Dpn1 and BsaI to remove the plasmid template and create overhangs for cloning (37°C for 1 hour). Ligation reactions were set-up according to the indications from NEB, with plasmid: insert ratios ranging from 1: 3 to 1: 7. Ligation was carried for 2 hours at room temperature, followed by transformation.

Gibson Assembly

Primers for the Gibson Assemblies were designed using SnapGene. 50 ng of plasmid backbone was mixed with the cloning inserts in a 1:4 ratio, according to the NEBioCalculator and NEBuilder® HiFi DNA Assembly Master Mix (NEB). Depending on the length of the inserted fragment, the reactions were carried for 15 minutes or 1 hour at 50 °C, with shorter fragments (hydrophobin sequences) being incubated for shorter time. Assembly products were transformed into DH10-beta chemicompetent cells.

Bacterial Transformation

Chemically competent *E. coli* cells were thawed on ice for 15 minutes. Cells were then transformed with 5 µL of cloning product or between 50 and 100 ng purified plasmid DNA. The DNA was mixed into the cells by gently flicking the tube. Cells were incubated on ice for 15 minutes, followed by a heat shock at 42 °C for 30 s. Cells were transferred on ice for 5 minutes and then supplemented to a total volume of 500 µL per tube with SOC Medium (NEB). Cells were incubated for 1 hour at 37 °C under shaking conditions (200 rpm). 100 µL of the bacteria were plated on kanamycin plates.

Colony PCR and sequencing

Q5 polymerase was used for colony PCR, with an additional initial of the bacterial colonies at 95 °C for 5 minutes. The success of cloning was then further validated by Macrogen Sanger sequencing. The Sanger sequencing data was analyzed using Benchling and aligned the sequencing traces to the *in sillico* assemblies of the vectors.

Protein Expression

Colonies of transformed BL21 cells were isolated and inoculated into 3 mL of LB media supplemented with 50 µg/mL kanamycin. An exponential culture was started using 500 µL of the overnight culture in a total volume of 50 mL of LB media supplemented with kanamycin. At an optical density (OD) equal to 0.5, protein expression was induced using IPTG to a final concentration of 400 µM. Cells were incubated for 3 hours in the presence of IPTG then pelleted at 4000 g for 20 minutes. Cell pellets were frozen for protein purification.

Protein Purification

Pelleted BL21 cells induced with IPTG were resuspended in 1X PBS buffer. The cells were sonicated for 15 x 30 seconds, with 30 second intervals between pulses. The mixture was centrifuged at 16000 g for 20 minutes and the supernatant was transferred to a clean tube. Then, proteins were extracted and purified using the HisPur Ni-NTA Resin kit (ThermoFischer Scientific), according to the manufacturer's specifications. An exception to the specification was that the protein extract was shaken for 1 hour in the resin in order to extract as much protein as possible from the sample.

Timelapse live cell imaging microscopy

BL21 cells expressing pET29 plasmid carrying recombinant protein-sfGFP fusions and untransformed controls were inoculated in LB media for overnight culture. The following day, exponential cultures were started. One condition used AB media (see annex 2 for exact composition of AB medium) supplemented with 50 µg/mL kanamycin, whereas the other one used AB medium without antibiotic for controls. After 3 hours of exponential growth, expression of the recombinant protein was induced by addition of IPTG (100 µM). After 1 hour of induction, cells were diluted to OD 0.01 and plated on AB-agarose pads containing IPTG and kanamycin. Controls of transformed BL21 cells were included on pads without IPTG, as well as untransformed BL21 cells in the absence of both kanamycin and IPTG. The pads were incubated at 37 °C under the Ti-Eclipse inverted microscope (Nikon) and the growth of cells, as well as the GFP fluorescence signal were monitored for 3 positions per pad and 42 time points per position. In the first hour of the experiment, images were recorded every 10 minutes, followed by image acquisition every 25 minutes in the hours following. The data generated by the experiment were analyzed by Nis-Elements Viewer (Nikon) and Fiji (ImageJ).

Yarrowia lipolytica growth curve generation

A single colony of *Yarrowia lipolytica* was inoculated in 3ml of YPD and incubated overnight at 30°C and 200 rpm. the OD was measured and diluted to 0.1 to initiate the growth experiment. The oil medium was prepared by adding 1.85 ml of oil to 48.15 ml of YP. The glucose medium was prepared by adding 6.25 ml of 40% glucose to 43.75 ml of YP. The concentrations of the carbon sources were 3.7% and 5% respectively, which were adjusted based on the moles of carbon and were equivalent. For more accuracy, two replicates of each culture were made. The Erlenmeyer shaking flasks were incubated at 30°C and 200 rpm and their ODs were measured every 2 hours for 60 hours. To account for the loss of data during night time, four additional cultures were prepared at the end of the first day of measurements and placed in the incubator to be measured in the following days.

Annex 1: Complete spreadsheet of all oligonucleotides used throughout the project.

Name	Sequence (5'-3')
Ecoli_pET28a_S2BE FWD	CTGCATGGTCTCCAACtcgagcaccaccaccaccac
Ecoli_pET28a_S2BE REV	CTGCATGGTCTCCTAtggctgccgcgcggcac
Ecoli_AcGFP FWD_pET28	CTGCATGGTCTCGcaTATGGTGTCCAAGGGCGCA
Ecoli_AcGFP REV_pET28	CTGCATGGTCTCGCCTGACCCGCCTTTATATAACTCGTCCATGCCATGT
Ecoli_MBSP1_FWD_pET28	CTGCATGGTCTCCCAGGTGGTTCTATGTCGGACCAATACCTT
Ecoli_MBSP1_REV_pET28	CTGCATGGTCTCCaGTTACGTGGAATCCTGGCC
Ecoli_HFBI FWD_pET28	GTGTCAGGTCTCCCAGGTGGTTCTATGTCTAACGGCAATGGTAACG
Ecoli_HFBI REV_pET28	GTGTCAGGTCTCCaGTTTTACGCCCCAACCGCTGT
Ecoli_HFBI_mutated FWD_pET28	GTCTCAGGTCTCGCAGGTGGTTCTATGTCCAACGGTAATGGCAAC
Ecoli_HFBI_mutated REV_pET28	GTCTCAGGTCTCGaGTTTCATGCCCCCACCGCTGT
Ecoli_HFBII FWD_pET28	AGTGTCGGTCTCGCAGGTGGTTCTATGAAGTTTTTTACTGCAGCGGC
Ecoli_HFBII REV_pET28	AGTGTCGGTCTCGaGTTTTATGCGGCCCCCACGGC
Ecoli_HFBII_mutated FWD_pET28	AATCCGGGTCTCGCAGGTGGTTCTATGAAATTCTTTACCGCAGCC
Ecoli_HFBII_mutated REV_pET28	AATCCGGGTCTCGaGTTTTAAGCGGCACCCACAGC
Ecoli_HGFI FWD_pET28	AACGCAGGTCTCCCAGGTGGTTCTATGTTTAGTAAACTTGCTATTTT
Ecoli_HGFI REV_pET28	AACGCAGGTCTCCaGTTTCATACGTTTACTGGAACG
Ecoli_HGFI_mutated_full FWD_pET28	GTACAGGGTCTCCCAGGTGGTTCTATGTTCAGCAAGTTAGCA
Ecoli_HGFI_mutated_full REV_pET28	GTACAGGGTCTCCaGTTTCATACATTAACTGGGACG
Ecoli_HGFI_mutated_half FWD_pET28	ACTGTCGGTCTCCCAGGTGGTTCTATGTTTTCCAAACTGGCG
Ecoli_HGFI_mutated_half REV_pET28	ACTGTCGGTCTCCaGTTTCATACGTTGACCGGAAC
pET29_linker_insert_fwd	CGCAATGGTCTCCTAGCATGGTTAGCAAAGGTGAA
pET29_linker_insert_rev	CGCAATGGTCTCCCACTTGAAGTGTATATCTCCTTC
pET29_link_vector_fwd	CCAGTTGGTCTCGTAAGTGGAGGTTCTGGAGGT
pET29_link_vector_rev	CCAGTTGGTCTCGTTGAAGTGTATATCTCCTTCTTAA
Ec HGFI FWD_pET29	CCAGTTGGTCTCGTCAATGTTTAGTAAACTTGCTATTTT
Ec HGFI REV_pET29	CCAGTTGGTCTCGCTTACGTTTACTGGAACGCA
Ec HGFI_mut_half FWD_pET29	CCAGTAGGTCTCCTCAAATGTTTTCCAAACTGGCGA
Ec HGFI_mut_half REV_pET29	CCAGTAGGTCTCCCTTATACGTTGACCGGAACGCA
Ec HGFI_mut_full FWD_pET29	CTTCCTGGTCTCCTCAAATGTTCAGCAAGTTAGCA
Ec HGFI_mut_full REV_pET29	CTTCCTGGTCTCCCTTATACATTAACTGGGACGGA
Ec MBSP1 FWD_pET29	CGTGTAGGTCTCCTCAAATGTCGGACCAATACCTTGACT
Ec MBSP1 REV_pET29	CGTGTAGGTCTCCCTTACGTGGAATCCTGGCCCGC
Ec HFBI FWD_pET29	CACACTGGTCTCGTCAAATGTCTAACGGCAATGGTAACG

Ec HFBI REV_pET29	CACACTGGTCTCGCTTACGCCCCAACCGCTGTTTG
Ec HFBI_mut FWD_pET29	GATCCTGGTCTCGTCAAATGTCCAACGGTAATGGCAAC
Ec HFBI_mut REV_pET29	GATCCTGGTCTCGCTTATGCCCCCACCGCTGTTTG
Ec HFBII FWD_pET29	ACTCTGGGTCTCGTCAAATGAAGTTTTTTACTGCAGCGGCGTTGTTTGC
Ec HFBII REV_pET29	ACTCTGGGTCTCGCTTATGCGGCCCCCACGGCGGC
Ec HFBII_mut FWD_pET29	TAGTGCGGTCTCGTCAAATGAAATTCTTTACCGCAGCCGCG
Ec HFBII_mut REV_pET29	TAGTGCGGTCTCGCTTAAGCGGCACCCACAGCGGC
linker_oligo_fwd_pET29	AGTGGAGGTTCTGGAGG
linker_oligo_rev_pET29	GCTACCTCCAGAACCTC
pET29_linker_insert_fwd	CGCAATGGTCTCCTAGCATGGTTAGCAAAGGTGAA
pET29_linker_insert_rev	CGCAATGGTCTCCCACTTGAAGTGTATATCTCCTTC
pET29_link_vector_fwd	CCAGTTGGTCTCGTAAGTGGAGGTTCTGGAGGT
pET29_link_vector_rev	CCAGTTGGTCTCGTTGAAGTGTATATCTCCTTCTTAA
Ec HGFI FWD	CCAGTTGGTCTCGTCAATGTTTAGTAAACTTGCTATTTT
Ec HGFI REV	CCAGTTGGTCTCGCTTACGTTTACTGGAACGCA
Ec HGFI_mut_half FWD	CCAGTAGGTCTCCTCAAATGTTTTCCAAACTGGCGA
Ec HGFI_mut_half REV	CCAGTAGGTCTCCCTTATACGTTGACCGGAACGCA
Ec HGFI_mut_full FWD	CTTCCTGGTCTCCTCAAATGTTCAGCAAGTTAGCA
Ec HGFI_mut_full REV	CTTCCTGGTCTCCCTTATACATTAACTGGGACGGA
Ec MBSP1 FWD	CGTGTAGGTCTCCTCAAATGTCGGACCAATACCTTGACT
Ec MBSP1 REV	CGTGTAGGTCTCCCTTACGTGGAATCCTGGCCCGC
Ec HFBI FWD	CACACTGGTCTCGTCAAATGTCTAACGGCAATGGTAACG
Ec HFBI REV	CACACTGGTCTCGCTTACGCCCCAACCGCTGTTTG
Ec HFBI_mut FWD	GATCCTGGTCTCGTCAAATGTCCAACGGTAATGGCAAC
Ec HFBI_mut REV	GATCCTGGTCTCGCTTATGCCCCCACCGCTGTTTG
Ec HFBII FWD	ACTCTGGGTCTCGTCAAATGAAGTTTTTTACTGCAGCGGCGTTGTTTGC
Ec HFBII REV	ACTCTGGGTCTCGCTTATGCGGCCCCCACGGCGGC
Ec HFBII_mut FWD	TAGTGCGGTCTCGTCAAATGAAATTCTTTACCGCAGCCGCG
Ec HFBII_mut REV	TAGTGCGGTCTCGCTTAAGCGGCACCCACAGCGGC
linker_oligo_fwd	AGTGGAGGTTCTGGAGG
linker_oligo_rev	GCTACCTCCAGAACCTC
pBEVY_fwd	caccaTCACCATTAATACCGAGCTCGAATTCGACAC
pBEFY_rev	GCAGTAAAAATTGAAGGAAATCTCATCCCGGGGAGTTGATTGTATGC
Sc_MF_fwd	GCATACAATCAACTCCCCGGGATGAGATTTCCTTCAATTTTTACTGCAG
Sc_MF_rev	TGAACCTCCAGAACCGCCACTTCTTTTATCCAAAGATACCCCTTCTTC
Sc_HFBI_fwd	AGTGGCGGTTCTGGAGGTTCATCTAACGGCAATGGTAACGTTTGtcc
Sc_HFBI_rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGCGccccaaccgctgtttg
Sc_HFBII_fwd	AGTGGCGGTTCTGGAGGTTCAAAGTTCTTCACTGCAGCAGCCC
Sc_HFBII_rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGAGCAGCACCAACAGCAGC
Sc_MBSP1_fwd	AGTGGCGGTTCTGGAGGTTCATCGGACCAATACCTTGACTTTG
Sc_MBSP1_rev	CGAGCTCGGTATTAATGGTGATGGTGATGGTGCGTGGAATCCTGGCCC

Annex 2: AB Medium Composition and preparation protocol.

<u>5xA (200 mL total):</u>

Ingredient	Overall ratio
$(NH_4)_2SO_4$	2 g/L (2 g)
$Na_2HPO_4 \times 2H_2O$	7.5 g/L (7.5 g)
KH_2PO_4	3 g/L (3 g)
NaCl	3 g/L (3 g)

Dissolve (the amount for 1L) in 200mL and autoclave

<u>1xB (800mL total):</u>
743,5 mL MilliQ water, autoclaved.
After autoclaving add:
+1 mL $CaCl_2$ (0.1M) +1 mL $MgCl_2$ (1M) +1 mL $FeCl_3$ (3mM)
+ 1mL Thiamine (10 mg/ml), +12.5 mL Uracil (2mg/ml)
+20 mL Casamino acids (10% m/v),
+ 20mL Glucose (20% m/v),

Add A into B to form 1L and mix well

<u>Stocks</u>
$CaCl_2$ 0.1 M, 1 mL per 1 L AB
0.147 g to 10 mL Milli-Q, can be autoclaved
$MgCl_2$ 1 M, 1 mL per 1 L AB
2,033g to 10 mL Milli-Q, can be autoclaved
$FeCl_3$ 0.003 M, 1 mL per 1 L AB
30 µL 1 M master-stock to 10 mL Milli-Q
rest of the stocks also in MQ, Casamino acids can be autoclaved, rest filter sterilized